

Project 1 (30%)

CIT1C18 - Computational Thinking

Introduction

This individual assignment aims to design a flowchart for a problem within the confines of the Temasek Polytechnic. Students are reminded to read through the entire assignment specifications to understand the complete requirements.

It is the start of a new semester, and the new intake of Year 1 students are new to the campus and unfamiliar with the locations around campus.

Your task is to design the program logic using a flowchart that will help the students navigate to a key location in Temasek Polytechnic while considering real-world factors that affect movement. You must consider **at least 2 real-world factors** in your flowchart design, such as but not limited to:

- **Weather Conditions:**
If it is raining, students might prefer covered walkways or indoor routes instead of open pathways. On good weather, the students might prefer the shortest route.
- **Crowd Level:**
During peak hours (e.g., lunchtime), certain walkways, canteens, and entry points might be congested.
- **Walking Difficulty / Mobility Needs:**
Some students may have difficulty walking long distances, using stairs, or moving quickly. They may prefer routes with lifts, ramps, sheltered rest points, or shorter walking distances.

Learning outcomes:

By the end of this assignment, you should be able to:

- Use necessary tools, such as a flowchart, to solve problems
- Describe and justify the problem-solving process

Note: The use of artificial intelligence tools, including but not limited to ChatGPT, is strictly prohibited for this project.

Task 1 – Location Selection and Visual Documentation (4%)

Walk through Temasek Polytechnic and identify TWO (2) locations that can be used as part of your problem.

Location 1 (Start Location):

Choose ONE (1) location from the options below as your starting location.

- McDonald's
- The Designer Pad (canteen in the School of Design)

Location 2 (End Location):

Choose ONE (1) location from the options below as the destination.

- The Short Circuit (canteen in the School of Engineering)
- The Flavours (canteen in the School of Informatics & IT)
- The Business Park (canteen in the School of Business)
- Sprout (canteen in the School of Humanities & Social Sciences)

Take a photo of each location and insert the **TWO (2) photos** into **Section A** of your report.

Important: The photos must be **clear and visually helpful**. They should show enough of the environment to help a user unfamiliar with the area understand their location and orientation.

What is a visually helpful photo?

A visually helpful photo is a photo that makes it easy for someone unfamiliar with the area to recognise where they are.

A visually helpful photo should:

- clearly show the location
- include unique landmarks
- help the user understand which direction they are facing
- not be too zoomed in, blurry, dark, or blocked

Task 2 – Flowchart Design (16%)

Imagine that you need to code out a program that will guide students from the **start location** to the **end location** in Task 1. (Example program below).

The program should minimally meet the following requirements:

- Display the start location.
- Prompt the user to input the destination location.
- Prompt the user to input details for real-world factors.
- Display the directions to the destination location.
- Run continuously until the user chooses to exit.
- There must be at least 3 unique routes to the end location.

Example Program:

```
You are at Mcdonald's in Temasek Polytechnic
Where do you want to go? (Enter 1 or 2)
1) The Short Circuit
2) Exit
1
Is it a peak period now? (yes/no): no
Do you have walking difficulty (yes/no): no

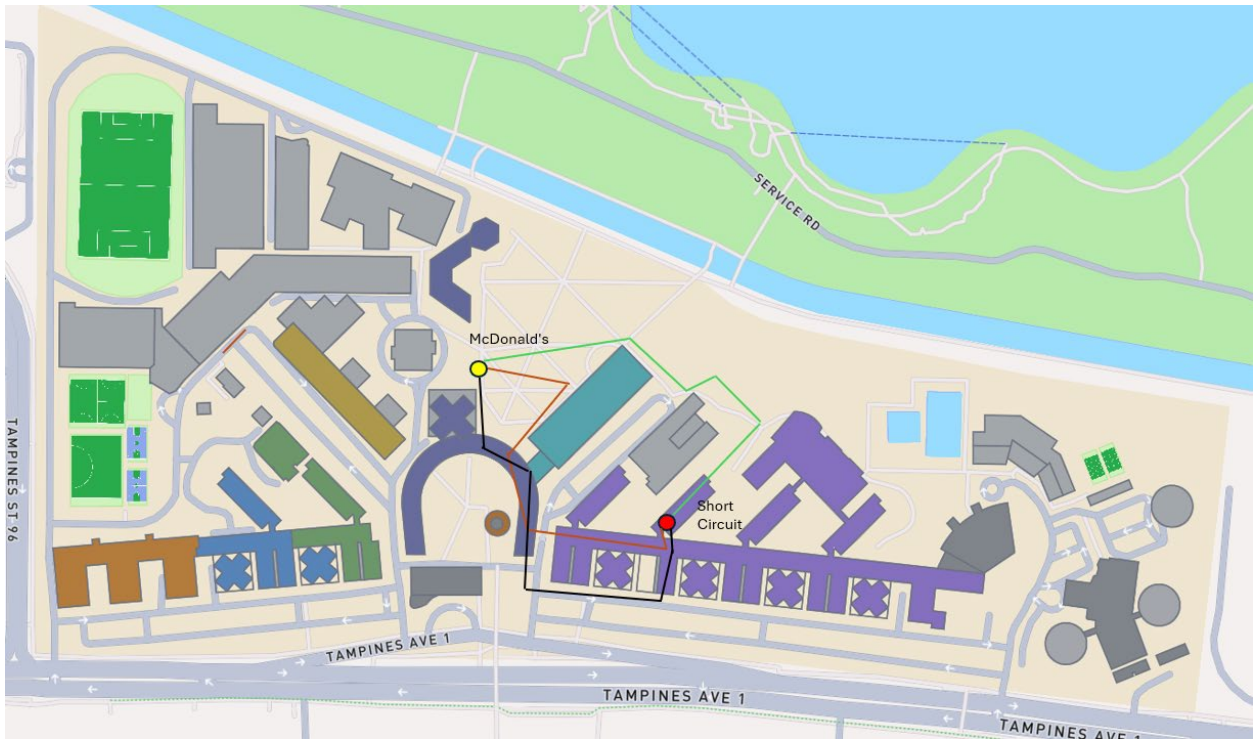
Here are the directions to The Short Circuit:

Turn to face The Designer Pad
Walk straight towards The Designer Pad
Turn to face Block 9
Walk straight towards Block 9
Climb [1] level of stairs up
.
.
.
.
Walk straight towards The Short Circuit
You will have arrived at the destination!

You are at Mcdonald's in Temasek Polytechnic
Where do you want to go? (Enter 1 or 2)
1) The Short Circuit
2) Exit
```

Based on your selected locations from Task 1:

- 1) Take a screenshot of the TP Campus Map and plot out the routes to the end location.
(TP Campus Map - <https://www.tp.edu.sg/about-tp/tp-campus-map.html>)
(Sample of plotted routes below, for illustration only)



2) **Create a flowchart** using draw.io to illustrate the program.

You must limit your flowchart to:

- no more than 60 blocks in total
(including start, end, decision, action, and input/output blocks)
- no more than 10 decision blocks
- no more than 10 input/output blocks

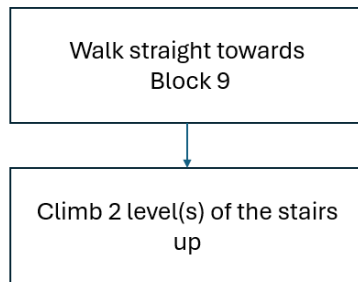
Here are the actions/directional instructions that you can use to guide the user:

- Walk straight towards [enter end point]
- Turn to face [enter location]
- Climb [x] level(s) of the stairs up
- Climb [x] level(s) of the stairs down
- Take [x] level(s) of the lift up
- Take [x] level(s) of the lift down

You may use decision blocks and other flowchart symbols between action blocks to control the flow. The instructions should make use of nearby landmarks or recognisable points along the route to guide the user progressively to the final destination. Avoid jumping directly to the final destination in a single step.

Note: Use **action blocks** for the directional instructions.

Example of directional instruction:



Insert the **map** and **flowchart** in **Section B** of your report.

(Can be a screenshot of the map and flowchart. Ensure that both are clear and visible.)

You can make use of the MapTemplate.pptx in LMS to help you with plotting the route.

Task 3 –Route Justification and Evaluation (10%)

After designing your flowchart, using the routes that you had mapped out to **Location 2**, explain the following:

- Why did you choose these routes to reach Location 2 instead of other possible options? In your explanation, describe the different routes you considered, including the one you initially planned to take. Then, discuss how your choice was influenced or changed after considering real-world factors.
- What made you decide that these were the most suitable routes? What factors did you consider. Support your decision with data.

Your justification and evaluation are not limited to just written explanations. You can include photos, images, or other forms of research data.

You must include relevant research data to support your decision.

You have a **500-word** limit.

In **Section C** of the report, include the explanation for your choice of route and the research data that may help clarify your decision-making process.

Submission

You are to create your Project 1 Word document report using the template ('COMT_AY2627_Project_1_Template.docx'). Submit your report and drawio file (.drawio) into LMS under the Assessment folder by **Week 6, Thursday, 28th May 2026, at 1700hrs (5 PM)**. You should name your document according to the following format:

StudentID_YourName_COMT_AY2627_Project_1.docx
StudentID_YourName_COMT_AY2627_Project_1.drawio

e.g.: 1234567D_Teo_Chu_Chu_COMT_AY2627_Project_1.docx
e.g.: 1234567D_Teo_Chu_Chu_COMT_AY2627_Project_1.drawio

File Submission and Backup Policy

When submitting assignments, each student is responsible for ensuring that they do not submit an empty zip file or a corrupted file.

Students must also ensure they are using a suitable device when working on assignments / projects, maintain proper backups of their work and do not cite 'laptop issues' as a reason for assignment extensions as such reasons will not be considered.

Penalty for Late Submission

Late within 1 day: 10% deduction from absolute mark given for the assignment e.g 75 marks (100 marks max) becomes 65 marks (deduct 10% of 100 marks)

Late > 1 day and up to 2 days: 20% deduction from absolute mark

Late > 2 days: 0 marks awarded

Project 1– Grading Criteria

The grading criteria for **Project 1 (30%)** will be based on the following:

Criteria	In Context	Performance Level				
		A (>=80%)	B (>=70%)	C (>=60%)	D (>=50%)	F (<50%)
		Excellent	Good	Average	Pass	Below Standard
Location Selection and Visual Documentation (4%)	Navigational Usefulness (4%) - Clarity - Usefulness of photos	Photos are clear, well-composed, and immediately help to recognise the location. Key features and landmarks are intentionally shown to support orientation.	Photos clearly show the location and include recognisable details, though could be better composed or include more helpful elements for full clarity	Photos show the location adequately, but may require some prior knowledge or assumptions to identify the place.	Photos capture the basic setting of the locations but lack clarity.	Non-submission, or photos are unclear, unusable, or lacks sufficient information to support any understanding of the location.

Flowchart Design (16%)	Logic (10%) - Correctness - Completeness - Flowchart Optimization	The flowchart is logical, correct, and efficient, with no missing steps or errors in the sequence. Flowchart design fully meets the program requirements. Flowchart includes multiple routes to the destination, taking into account multiple real-world factors that affect movement, and adjusts the routes accordingly. All routes are optimised. The flowchart contains no unnecessary repeated steps. While there may be some necessary repeated steps to maintain clarity or ensure the process is	The flowchart is logical and correct. Flowchart design fully meets the program requirements. Flowchart includes multiple routes to the destination, taking into account multiple real-world factors. The flowchart contains some unnecessary repeated steps that do not complicate the overall solution.	The flowchart is mostly logical and correct. Flowchart design meets most of the program requirements. Flowchart includes multiple routes to the destination, taking into account one real-world factor. The flowchart contains some repeated steps, which affect the clarity of the solution.	The flowchart is somewhat logical and correct. Flowchart design meets some of the program requirements. Flowchart includes primary route to the destination. The flowchart contains many repeated steps, complicating the solution unnecessarily.	Non-submission, or incomplete flowchart, or flowchart that does not meet the program requirements.
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		followed correctly, the overall solution is clear and concise.				
	Flow Chart Conventions (3%)	All standard flowchart symbols are used correctly, and the flow is clear and easy to follow. The organisation is neat, clear, and well-structured throughout, making the flowchart easy to follow.	All standard flowchart symbols are correctly used, with a clear and logical flow that is mostly easy to follow. The organisation is generally neat and clear, making the flowchart mostly easy to follow.	Most of the standard flowchart symbols are correctly used. The organisation is somewhat clear but still lacks neatness and clarity, making the flowchart difficult to follow.	Some of the standard flowchart symbols are correctly used. Poor organisation, making the flowchart difficult to follow.	The flowchart does not use appropriate symbols, and the layout is disorganised. It is difficult to understand or completely unclear.
	Constraints (3%) - Keeping to the constraints and limitations specified	The flowchart strictly adheres to all constraints (no more than 60 blocks in total, 10 decision blocks and 10 input/output blocks). Fully adheres to the given directional instruction format throughout.	The flowchart strictly adheres to all constraints (no more than 60 blocks in total, 10 decision blocks and 10 input/output blocks). Mostly adheres to the given directional instruction format,	The flowchart adheres to at least two constraints (no more than 60 blocks in total, 10 decision blocks and 10 input/output blocks). Partially adheres to the given directional instruction format.	The flowchart adheres to at least one of the constraints (no more than 60 blocks in total, 10 decision blocks and 10 input/output blocks). Minimally adheres to the given	The flowchart shows little to no consideration of the constraints.

			with minor inconsistencies.		directional instruction format.	
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Route Justification and Evaluation (10%)	Route Justification and Evaluation (10%) - Clarity of justification - Thoroughness of evaluation - Evidence of research	Clear, logical, and well-supported justification for the choice of routes, with strong reasoning for why these specific routes were chosen over possible alternatives. Thorough and detailed evaluation of all relevant factors, showing a deep understanding of how each one impacts the route decisions. All research evidence presented strongly supports the justification of the route choices. The research data are highly relevant, well integrated and strongly reinforce the evaluation.	Clear and reasonable justification for the choice of routes, but more depth or detail could enhance the explanation. Detailed evaluation of most relevant factors, showing a good understanding of how they impact the route decisions. Multiple pieces of research evidence are provided, with clear links to support the justification of the route choices. The research data are relevant and reinforce the evaluation.	Some justification for the choice of routes is provided, but the reasoning lacks clarity or depth, and further exploration of key aspects. Evaluation touches on some relevant factors, but the analysis is basic and lacks depth. Relevant research evidence is used, but may not consistently support the route choices, with gaps in explanation or integration into the overall evaluation.	Vague explanation provided of why the routes were chosen. Evaluation mentions a few factors but lacks meaningful analysis or explanation of their impact on route decisions. Research evidence is limited and weakly supports the route choices, with poor integration into the overall justification.	No clear justification is provided, or reasoning is completely absent. No evaluation of key factors. The analysis is missing or entirely superficial. No research evidence to support the justification, and the explanation is vague or superficial.
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